

Linear combinations, span, and basis vectors

CHAPTER 2

Linear combinations, span and basis

What can you build, and what can you reach?

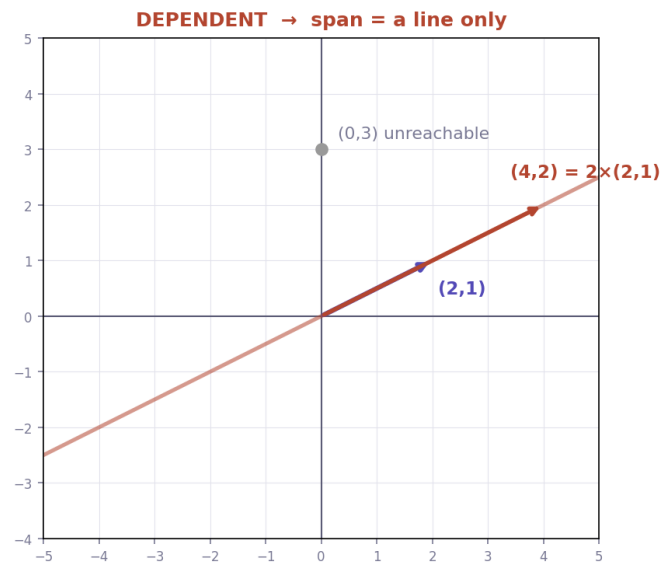
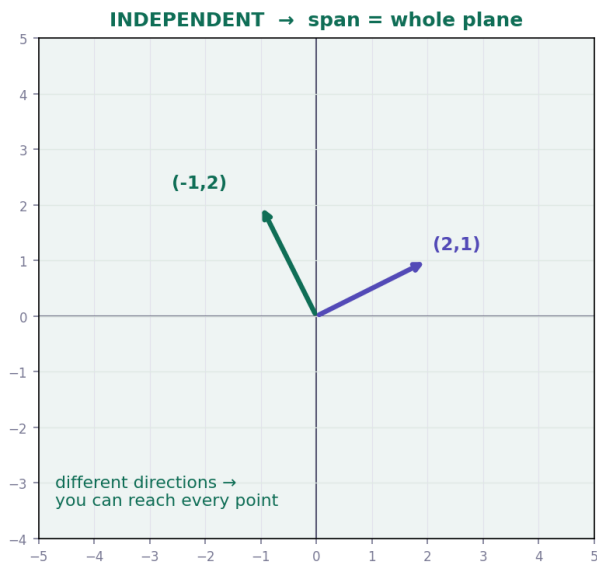
THE IDEA

Take some vectors, **scale each one** by a number, then **add the results**. That is a **linear combination**. The set of **every point** you can reach this way is the **span**.

Term	Meaning
Scalar	An ordinary number that stretches or shrinks a vector
Linear combination	$c_1v_1 + c_2v_2 + \dots$ — scale each, then add
Span	Every point reachable by linear combinations of the given vectors
Basis	A minimal set of vectors whose span is the whole space (e.g. $\hat{i}$ and $\hat{j}$ in 2D)

WHEN SPAN COLLAPSES

Two vectors pointing in **different directions** span the whole plane. But if one is a scaled copy of the other — **parallel** — the span shrinks to a single line, and most of the plane becomes unreachable. That is the situation called **linearly dependent**.



INDEPENDENCE VS DEPENDENCE

Case	Test	Span in 2D
Independent	No vector is a combination of the others	Whole plane
Dependent	One is a scaled copy / combination of the others	A line (or a point)

Quick check in 2D: are they multiples of each other? $(2, 1)$ and $(6, 3)$ are dependent because $(6, 3) = 3 \times (2, 1)$.

PARAMETRIC LINES — THE SAME IDEA

A line written $(x, y) = (1, 2) + t(3, -1)$ has two different jobs inside it: **(1, 2) is the anchor** (used once, it fixes where the line sits) and **(3, -1) is the direction** (multiplied by t , repeated at every scale). The dial t slides you along the line: $t = 0 \rightarrow (1, 2)$, $t = 2 \rightarrow (7, 0)$.

WORKED EXAMPLES

CHAPTER 2 — WORKED EXAMPLES

LINEAR COMBINATION

$$v = (1, 2) \quad w = (3, -1)$$

$$2v = (2, 4) \quad 3w = (9, -3)$$

$$2v + 3w = (2+9, 4+(-3)) = (11, 1)$$

FINDING UNKNOWN SCALARS

$$\text{find } c_1, c_2 \text{ with } c_1(1,2) + c_2(2,-1) = (4,3)$$

split into components :

$$x : c_1 + 2c_2 = 4$$

$$y : 2c_1 - c_2 = 3$$

$$\text{solve} \rightarrow c_1 = 2, c_2 = 1$$

$$\text{check : } 2(1,2) + 1(2,-1) = (2,4) + (2,-1) = (4,3) \quad \checkmark$$

SPAN TEST

$$u = (1,0), v = (2,0) \rightarrow \text{both have } y = 0$$

$$\text{any combination} = (c_1 + 2c_2, 0) \rightarrow y \text{ always } 0$$

span = the x-axis only. (0,3) is NOT reachable

Why this matters for ML: if one feature of your dataset is a linear combination of others, it carries no new information — it lies inside the span of what you already have. That is exactly what "redundant" or "collinear" features mean.

TEST — Chapter 2

1. $v = (1, 2)$, $w = (3, -1)$. Compute $2v + 3w$. $\rightarrow (11, 1)$

2. $a = (2, -4)$, $b = (-5, 1)$. Compute $-3a + 2b$. $\rightarrow (-16, 14)$

3. Find c_1, c_2 with $c_1(1, 2) + c_2(2, -1) = (4, 3)$. $\rightarrow c_1 = 2, c_2 = 1$

FORMULAS FROM THIS CHAPTER

Linear combination	$c_1v_1 + c_2v_2 + \dots$
Dependent (2D)	one vector is a multiple of the other
Span (dependent)	a line, not the plane
Parametric line	point = anchor + $t \cdot$ direction